

Statistically-Validated Econophysics Edges in Crypto Derivatives

A review of the FQ signal engine and its anti-overfitting methodology

Status: preprint / working draft — reproducible from source; to be hardened (data appendix, DOIs) before journal submission. ·

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ABSTRACT

We present **FQ**, a directional signal engine for cryptocurrency perpetual futures built on the **econophysics** program and validated with the modern anti-overfitting standard of quantitative finance — the **Deflated Sharpe Ratio**, **CPCV** and **PBO** (López de Prado). Unlike most retail systems — which report no multiple-testing correction — FQ subjects **every** candidate edge to an explicit statistical gate and measures performance *forward* before raising conviction. This review documents the validation methodology; the edges that survive — signed order-flow (CVD), order-flow persistence, regime detection via time-irreversibility (KL), distance to the volume-profile Point of Control, and the funding-rate percentile as a directional bias; the honest separation between a heuristic conviction layer and the causally-validated edge; a cross-asset extension via a data/venue decoupling; and an explicit graveyard of refuted ideas. The operating principle is singular: **measure or perish**.

1. Introduction

Econophysics — statistical-physics methods applied to markets (Mantegna & Stanley, 2000) — produced a robust core: heavy tails, power laws, square-root impact, long memory of order flow. The challenge is not a shortage of phenomena but separating the **real, samplable, operationalizable edge** from the **overfitting artifact** and from *physics envy*. FQ takes a strict stance: an idea enters the engine only if (a) it passes a gate that **deflates** by how many things were tried, (b) it **adds** within the already-confirmed edges (orthogonality), and (c) it is measured *forward* before any capital or conviction decision. This review maps what survived.

2. Methodology: the validation gate

The system's core is not new data — it is **validation**. Under multiple testing, ~20 configurations guarantee a spurious "5% significant" result. The **Deflated Sharpe Ratio** (Bailey & López de Prado, 2014) corrects the Sharpe for the number of trials and non-normality (skew/kurtosis), requiring **DSR > 0.95**. It is complemented by **CPCV** (combinatorial purged cross-validation with embargo, for leak-free out-of-sample estimation) and **PBO** (probability of backtest overfitting; Bailey et al., 2017). FQ implements all three in `tools/validation_gate.py`. It does not generate alpha — it **certifies** it.

3. The validated edges

3.1 Signed order-flow (CVD)

Cumulative Volume Delta gives the **sign** of flow; its basis is the **square-root impact law** ($\sigma \cdot \sqrt{(Q/V)}$), confirmed in crypto (Donier & Bonart, 2015; Sato & Kanazawa, *PRL* 2025). The CVD-confirmed subset (imbalance ≥ 0.50) yields **+0.27R (SOL)** / **+0.34R (BTC)** over 5 years of tick data, **DSR ✓**. Causal and free (Binance `aggTrades`). A +1.47R result at $n=17$ was discarded by the gate as a small-sample mirage.

3.2 Order-flow persistence (long memory)

Large orders execute in pieces → **positive autocorrelation of flow sign** (Lillo, Mike & Farmer, 2005; Bouchaud, Farmer & Lillo, 2009). FQ measures lag-1 autocorrelation (F2). In BTC the premium tier (CVD ✓ & F2 ✓) reaches **DSR 0.995**. **Honest finding**: F2 is an idiosyncratic per-symbol confirmer (pays in 4/6; negative in ETH), **not** a universal scaling law — the "scales with institutional-ness" hypothesis was **refuted** (corr = -0.19).

3.3 Regime via time-irreversibility (KL)

The thermodynamic arrow of time: KL divergence between the degree distributions of the **horizontal visibility graph** forward vs. backward measures distance from equilibrium = entropy production (Kawai, Parrondo & Van den Broeck, 2007; Lacasa et al., 2012). **No free parameters**. The edge lives in **low** irreversibility (mean-reverting): **BTC +0.348R DSR 0.999; SOL +0.225R DSR 0.950**, monotone by quartile and **cross-symbol**.

3.4 Distance to the volume-profile Point of Control

The newest piece. The prior-day volume profile defines the **POC** and *Value Area*. The hypothesis — "do not trade in yesterday's chop; far from the POC = trend" — **passes the cross-symbol gate**: a 5-crypto pool (n=5162) shows uplift +0.121, **orthogonal to KL** (within-KL +0.272), **CPCV OOS +0.111 (93% of paths positive)**, **PBO 0.17**. It holds in 4/5; **BNB is the measured exception** (far<near) and is excluded (Appendix B).

3.5 Funding-rate percentile (directional)

The newest piece and the **strongest gate result in the program**. Perpetual funding is the carry that anchors the perp to spot; high funding = crowded longs (BIS WP1087 links high carry to crash risk; retail trend-chasers inflate it). **Key finding**: the **raw level does not inform** (PBO 0.75), but the **percentile relative to the symbol's own 90-day history does**. It is directional and asymmetric: **LONG on cold funding** (pctl ≤ 0.5) → +0.173R vs +0.121 base (n=1538), **DSR 1.000, CPCV OOS +0.028 (80% of paths > 0), PBO 0.04**; **SHORT on hot funding** (pctl ≥ 0.7) → +0.224R vs +0.156, **DSR 1.000, CPCV 100% of paths, PBO 0.00 — the best gate result in the program**. The gradient is clean and monotone: longs decay as funding heats (+0.175 → +0.095) and shorts mirror it. It is wired along the **same path as CVD**: gate ✓ → dormant (sealed as a regime tag) → forward (judged by *by_funding* in the ledger) → product (directional conviction boost). Cross-venue note: validated on Binance history; live, each venue is compared to its own 90-day history (same relative construct).

4. Conviction vs. validated edge: the honest distinction

FQ separates two layers that most systems conflate. **Conviction** (P_master) is a structured heuristic — golden ratio ϕ weighted by confluence, ICT structural concepts (sweeps, order blocks, fair-value gaps, session killzones), a learned memory κ , and an *emergent time* factor σ_τ from a Monte-Carlo of price-path trajectories. It is principled but **not** a certified causal edge. The **validated edge** is the overlay that passed DSR/CPCV/PBO (CVD, F2, KL, POC-distance), measured forward. Conflating the two is the origin of hype: the heuristic **gates and prioritizes**; the validated edge **decides**.

5. Cross-asset extension: the data/venue decoupling

The edge is a property of the **asset**, not the venue. For traditional markets (gold, NASDAQ, S&P, oil, silver), FQ validates on a deep, free historical record and executes on the exchange's perpetual. Price-based features transfer (KL, base structure, the NY-calibrated ICT layer); order-flow (CVD/F2) does **not**. Preliminary: the engine digests traditional-market OHLCV and produces cubes with a positive base edge; POC-distance shows the **same direction** as in crypto, though the full gate remains under-powered (low n) pending a larger harvest.

6. Results and graveyard

Survives and is wired: CVD (DSR ✓), F2 (DSR ✓, BTC), KL (DSR ✓ cross-symbol), POC-distance (gate ✓; Appendix B), **directional funding percentile** (gate ✓; the best in the program, §3.5). **Passes but redundant:** F1 (impact residual; within-CVD negative). **Measured and refuted in the same sweep (do not re-chase):** the *breadth*/alt-season gate for alt longs (uplift +0.002, **PBO 1.00** — the macro decoupling does not descend to a 5-minute gate; horizon mismatch); on-chain valuation metrics (NUPL) do not clear the slow-regime protocol ($P = 0.941 < 0.95$). **Refuted / unfalsifiable (not coded):** Baaquie quantum finance, Sornette LPPL, Khrennikov quantum-like markets, critical-slowness, rough volatility (Cont & Das, 2022). What the gate killed is written down.

7. Discussion and limitations

Cube figures are **gross** (pre-cost) and in-sample to the harvest period; validated with CPCV/PBO, but the final judge is **forward**. FQ measures forward in *paper* (0% capital) and requires ≥ 30 –50 fills + uplift + DSR before client-facing conviction. The traditional-market extension is n-limited. None of these is fatal; all are measured.

8. Conclusion

FQ is not a system of promises: it is a **program for validating econophysics edges** with an honest record of what happened, on what data, over what horizon. Its contribution is not a secret alpha but a **reproducible methodology** — take the robust core of statistical market physics and run it through the field's most demanding anti-overfitting gate, *forward before belief*. In a domain saturated with overfitting and hype, that is, in itself, the result.

Appendix A — Disciplinary lineage

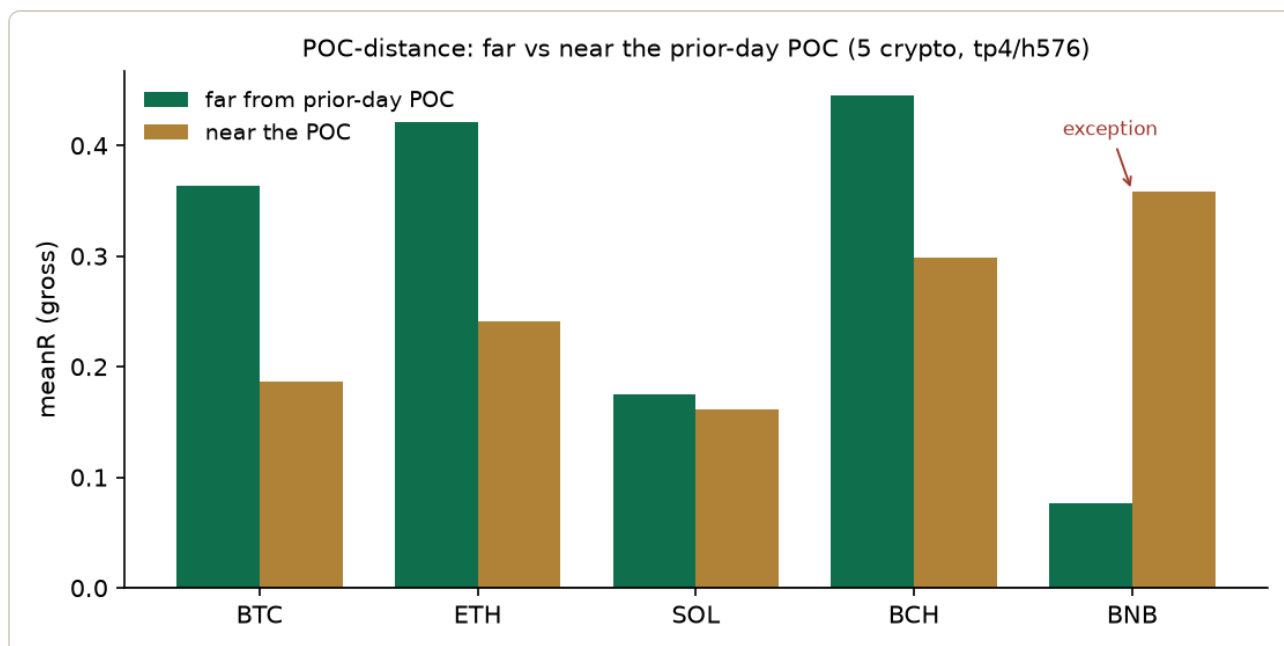
The natural profile is a **physicist or applied mathematician turned quant**. FQ's components map to real founders: DSR/CPCV/PBO → **López de Prado** (Cornell ORIE / ADIA); square-root impact, order-flow → **Bouchaud** (CFM / École Polytechnique); long-memory order-splitting → **Lillo, Mike, Farmer, Bouchaud**; irreversibility & visibility graphs → **Parrondo, Van den Broeck, Lacasa**; econophysics → **Mantegna & Stanley**. Taught at the world's top programs (Cornell, Oxford, Princeton, CMU, ETH). The twist: done self-taught from Mexico City, with a live, forward-validated system rather than a paper thesis.

Appendix B — Reproducible results (POC-distance)

Generated with `python tools/reproduce_gate_results.py` over the 5 crypto cubes (tp4/h576, gross, using the *same gate_poc_distance* that validates in production).

SYMBOL	N	FAR	NEAR	UPLIFT
BTC	1005	+0.364	+0.187	+0.177
ETH	1155	+0.421	+0.241	+0.179
SOL	976	+0.175	+0.161	+0.014
BCH	1503	+0.445	+0.299	+0.146
BNB	523	+0.077	+0.358	−0.281 (exception)

Pooled (n=5162): uplift +0.121 · **DSR 1.000** · orthogonal to KL (within-KL +0.272) · **CPCV OOS +0.111 (93% of paths >0)** · **PBO 0.17** → **PASSES**. 4/5 follow far>near; BNB is the measured exception and is excluded.



References *(preliminary — to be completed)*

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